MIL-S-46989A(MU)
18 DECEMBER 1963
SUPERSEDING
MIL-S-46989(MU)
16 JULY 1963

MILITARY SPECIFICATION

SLEEVES, ENCASEMENT, NYLON, KNITTED

1. SCOPE

1.1 This specification covers the quality assurance requirements, and special requirements not covered by the drawings, for the processing of knitted tubular sleeving designated as Encasement Sleeves (see 6.6).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-10025 —Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for

STANDARDS

MILITARY

MIL-STD-105D —Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8845230—Sleeve, Encasement
8845231—Sleeve, Encasement
8845252—Sleeve, Encasement

PUBLICATIONS

ORDNANCE CORPS

IEL-8873410—Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawing.

3.2 Encasement sleeves. The encasement sleeves shall comply with all the requirements specified on Drawings (dwg.) 8845230, 8845231 or 8845252, as applicable, and with all requirements specified in this specification.

3.3 Construction. The sleeving shall be knitted using a nonrunning lockstitch and shall have 64 to 68 threads. A 6 ply/210 denier/34 filament thread shall be used as the base yarn with a single end of 210 denier used for the locking stitch. The filaments shall be of nylon, Du Pont type 330, 380 or equal. When the sleeving is unrestrained, there shall be a minimum of 11 double stitches per inch across the width and a minimum of 12 stitches, or loops, per inch along the edge (see 6.4).

3.4 Defects.

3.4.1 *Knots.* Knots, or knots with up to 3 skipped stitches shall not be considered as defects.

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3.4.2 *Repairs.* An overlocking stitch shall be used for repairing. Repair stitches shall abut each other (see 6.5).

3.4.2.1 *Longitudinal.* Longitudinal repairs shall not exceed 4 inches in length.

3.4.2.2 *Circumferential.* Circumferential repairs or butt seams shall not exceed 3 in a 70 foot sleeve or 5 in a 140 foot sleeve.

3.5 **Workmanship.** The sleeving shall be constructed using the best knitting practices. Individual sleeves shall be free of oil and grease. Dirt smudges from handling shall not be considered objectionable.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier shall be responsible for the performance of inspection to assure the quality required by this specification.

4.1.1 *Submission of product.* At the time a completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all technical and quality assurance requirements.
(b) Drawing and specification number and date, together with an identification and date of changes.
(c) A certificate of compliance from the material supplier.
(d) Number of items in the lot.
(e) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certification submitted shall be substantiated by evidence of the agents authority to bind his principal. Substantiation of an agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using this detail specification, the applicable drawings and other contractual documents, the Government inspector shall verify all operations and inspection functions performed by the contractor. Verification will be performed to the extent necessary to assure compliance with contractual requirements.

4.2 Initial production sample.

4.2.1 *Submission.* Prior to the start of production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.7) for evaluation in accordance with the requirements of 4.3.1 and 4.3.2. The initial production sample shall consist of 10 each encasement sleeve under contract. The sample, or samples, shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production. The sample, or samples, shall be accompanied by documents which attest that the material supplied is the material required by the contract. An initial production sample, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, specification or source of supply as to significantly affect product uniformity, as determined by the Government. Prior to submission, the contractor shall inspect the sample, or samples, to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 *Inspections to be performed.* The sleeves will be subjected by the Government to the examinations of 4.3.1 and 4.3.2.

4.2.3 *Rejection.* If any sleeve fails to comply with any applicable requirement, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample, or portion thereof.

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Until an initial production sample is accepted, the contractor is, in no way autnorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 **Inspection provisions.**

4.3.1 *Examination.* The sampling plan and procedure for the following classification of defects shall be in accordance with Inspection Level I, table I of Standard MIL–STD–105D. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.1.1 Encasement sleeve (see dwg. 8845230 or 8845231, as applicable).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.3).*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Length, minimum (min.)	Gage	01001
102.	Width	Gage	01002
103.	Unrepaired defect	Visual	01003
Minor:	AQL 1.00 percent		
201.	Excess of circumferential seams	Visual	01004
202.	Evidence of poor workmanship (see 3.5)	Visual	01005

4.3.2 **Testing.**

4.3.2.1 Initial production sample (see 4.2.1). The samples submitted as the initial production sample shall be examined as specified below after having been examined for all other drawing requirements. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.2.1.1 Number of stitches (see 3.3 and 4.4.1)—Code No. 02001.

4.3.2.1.2 Non-running stitch (see 3.3 and 4.4.2)—Code No. 02002.

4.3.2.2 *Production testing.*

4.3.2.2.1 Number of stitches (see 3.3)—major defect—Code No. 02001. One encasement sleeve from each 100 submitted for acceptance shall be subjected to this examination which shall be performed as specified in 4.4.1. Failure of 1 unit in meeting either minimum stitch requirement shall reject the entire quantity submitted subject to 100 percent inspection and resubmission to Government inspection.

4.3.2.2.2 Non-running stitch (see 3.3)—major defect—Code No. 02002. A 6 inch length of sleeving shall be cut from 2 sleeves of each lot submitted for acceptance for the performance of this test. The cutting of the sample shall not reduce the sleeve length to less than 6 inches longer than the minimum length requirement.) Testing shall be as specified in 4.4.2. Failure of either sample in meeting the requirement shall reject the lot.

4.3.3 *Inspection equipment.* Index of Inspection Equipment Number IEL–8873410 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.3.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.3.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

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4.3.3.3 *Government use of contractor's inspection equipment.* The contractor shall as required make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Operation of the equipment and verification of its accuracy shall be supplied by the contractor, as required by the Government in performance of its examination or test.

4.4 **Test methods and procedures.**

4.4.1 *Number of stitches.* The sample shall be measured for compliance with the sleeving width requirement. To count the double stitches across the width, the sample shall be so placed that light reflection shows seemingly straight lines parallel to the edge. An inch graduation of a scale shall be located over one of the lines and the number of lines in an inch counted. The same procedure shall be used for counting stitches on the edge. The scale shall be located at the end of a stitch or loop. Only one line of stitches shall be counted.

4.4.2 *Non-running stitch.* The sample segment shall be cut for approximately 2 inches between double stitch lines starting not less than 1 inch from an end. The forefinger of each hand shall be inserted into the cut. Full force shall then be exerted against the edges of the cut perpendicular to the double stitch lines. If stitches unlock and thereby increase the length of the cut, the sleeving is defective.

5. PREPARATION FOR DELIVERY

5.1 **Packing and marking.** The encasement sleeves shall be packed and marked in accordance with Specification MIL–P–10025.

6. NOTES

6.1 **Ordering data.** Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 **Inspection equipment.** The contractor shall design inspection equipment as required by the referenced inspection equipment lists in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design". Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Ordnance designs.* Ordnance designs are detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an IEL references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirement of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.5 *IEL codes.* The inspection equipment referenced in paragraphs 6.3.1, 6.3.2, 6.3.3, and 6.3.4 will be designated in the IEL by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor design responsibility on contractor format.
OD —Ordnance design.
ODMU —Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification control drawing.

6.3.6 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Knitting equipment. To the best knowledge of the Government a Raschel 24 gauge (24 needles in 2 inches) double needle bar knitting machine is the only equipment capable of manufacturing the sleeving.

6.5 Thread for repairs. Coates and Clark's Nylob S–138 thread, or equal, should be used for repairing.

6.6 Intended use. It is intended that the items of this specification will be used in Linear Demolition Charges, M58, M68, M96 and M97 for Demolition Kits, Projected Charge, M125, M126, M173 and M174.

6.7 Submission of initial production sample. Instructions as to the location for the evaluation of the initial production sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–NI.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. 1375–A671

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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